
Response to Reviewer 4u3P:

Weakness 1: Thank you for your valuable feedback. We have improved our research by comparing it with the DKD method. Our findings show that DKD is not appropriate for knowledge distillation in semantic segmentation. In our paper, we cited DKD in line 098, mentioning that *DKD (Zhao et al., 2022) indicates that Vanilla KD [7] is...* Here, we have outlined the main distinctions between our approach and DKD:

Task Differences with DKD: Originally designed for image classification and object detection tasks, DKD encounters challenges when utilized for knowledge distillation in semantic segmentation. To implement DKD for knowledge distillation in this context, the explicit pixel-wise separation of target and non-target classes is required, utilizing Ground Truth segmentation maps. However, in semantic segmentation tasks, the predicted segmentation maps from the model may vary in size compared to the Ground Truth segmentation maps. Additionally, DKD does not account for the presence of background class pixels in semantic segmentation, further complicating the separation of target and non-target pixels.

Overcoming DKD limitations: Our proposed method overcomes the limitations of DKD by dividing each channel into multiple sub-regions and implementing region-wise knowledge distillation, eliminating the requirement for explicit pixel-wise separation based on Ground Truth segmentation maps. Through the analysis of Equation (3) in the submitted paper, it is evident that the use of region-wise knowledge distillation negates the necessity for explicit separation of target and non-target regions.

Focusing on spatial relationships: Our method emphasizes spatial relationships between pixels within channels for knowledge distillation in semantic segmentation tasks, while DKD focuses on the relationship between classes within pixels. DKD calculates softmax along the channel dimension to reflect class relationships, whereas our method calculates softmax within divided $K \times K$ regions to capture spatial relationships between adjacent pixels.

DKD introduced only to explain the effectiveness of our method: The concept of DKD is only utilized to elucidate the effectiveness of our method, but it is not directly implemented. Our approach implicitly decouples the channel into target and non-target regions, contributing to its efficiency and effectiveness. In the submitted paper, for the sake of intuitive explanation, we refer to regions containing some or all target class pixels as target regions, and regions containing only non-target class pixels as non-target regions.